Mechanistic Interpretability of Commonsense Reasoning Failures in LLMs: A Paired Behavioral–Intervention Study

Mido Sang and Idea-Explorer

Abstract

We study a central question in LLM reliability: when commonsense reasoning fails, is the main issue missing knowledge or a breakdown in integrating question goals with candidate actions? We present a paired behavioral–mechanistic protocol that evaluates the same COMMONSENSEQA examples under DIRECT and COT prompting, then applies causal-style interventions in a local open-model proxy. On 120 validation items, GPT-4.1 reaches 90.0% accuracy with DIRECT prompting and 89.2% with COT prompting, with low bidirectional failure flips (DIRECT →COT: 2.5%; COT →DIRECT: 1.7%). A paired McNemar exact test shows no significant difference ($p = 1.0$). We further run layer-wise residual ablations in DISTILGPT2 on matched success/failure examples and test whether a hypothesized target layer has stronger causal effect than control layers. The targeted effect is not statistically significant (paired t-test $p = 0.853$, Wilcoxon $p = 0.546$, Cohen's $d = 0.038$). These results do not support a strong claim that COT broadly degrades commonsense performance in this sampled setting, and they provide only weak evidence for localized coherence circuits with this small proxy model. The main practical takeaway is methodological: robust parsing and paired evaluation design are necessary before drawing mechanistic conclusions from apparent reasoning failures.

1 Introduction

Commonsense failures in language models still appear in tasks that humans solve reliably, and these errors can propagate into real downstream decisions. Our main question is simple: are these failures mostly missing-knowledge errors, or do they reflect internal breakdowns in integrating question goals with plausible actions?

This distinction matters for intervention design. If the issue is knowledge coverage, then more data and retrieval should dominate. If the issue is integration, then debugging should focus on circuits, representations, and causal pathways. Recent work on toxic chain-of-thought behavior suggests that reasoning traces can hurt correctness in some settings [Li et al., 2024], while other mechanistic studies show that multi-step reasoning can rely on identifiable architectural motifs [Hou et al., 2023, Cabannes et al., 2024]. Yet most evaluations do not connect paired behavioral flips and intervention-based evidence on matched examples.

We address this gap with a compact paired protocol. We first run the same COMMONSENSEQA items with DIRECT and COT prompting on GPT-4.1, then measure per-item flips and paired significance. We next score goal-action coherence on a subset, and finally run layer-level ablations in DISTILGPT2 as an intervention proxy for causal localization. Figure 1 previews the main behavioral finding: accuracy is nearly unchanged between prompt conditions.

Quantitatively, DIRECT reaches 90.0% and COT 89.2% on 120 sampled COMMONSENSEQA validation examples; the paired difference is not significant (McNemar exact $p = 1.0$). Discordant flips

39th Conference on Neural Information Processing Systems (NeurIPS 2025).

are rare in both directions. In the intervention proxy, we find no significant target-layer effect over control layers.

We make four contributions:

- We propose a reproducible paired behavioral–mechanistic protocol for commonsense failure analysis.
- We conduct controlled DIRECT vs. COT comparisons on identical items with paired statistical tests.
- We complement behavior-level evaluation with intervention-based layer ablations on matched success/failure subsets.
- We provide evidence that strong CoT-specific degradation claims are not supported in this sampled setting, and we identify concrete limitations for next-step mechanistic studies.

Paper organization. Section 2 reviews prior work, section 3 describes the protocol, section 4 reports quantitative findings, and section 5 discusses implications and limitations.

2 Related Work

Behavioral failures under chain-of-thought prompting. Li et al. identify *toxic CoT* cases where chain-of-thought flips a correct direct answer to an incorrect one, and they trace part of this behavior to reduced question-signal retention in rationale generation [Li et al., 2024]. Their results motivate our paired flip-rate analysis. Unlike prior work emphasizing mitigation, we focus on testing whether the effect appears in our sampled COMMONSENSEQA protocol and whether it aligns with intervention evidence.

Mechanistic studies of reasoning circuits. Several studies report structured internal mechanisms for reasoning. Hou et al. propose MechanisticProbe and recover tree-like reasoning signals from attention patterns [Hou et al., 2023]. Cabannes et al. isolate iteration heads that support iterative computation in chain-of-thought settings [Cabannes et al., 2024]. These findings motivate our hypothesis that commonsense errors may involve coherence integration failures rather than pure knowledge absence.

Causal intervention and representation-level analyses. Recent work applies intervention tools such as attribution tracing and causal pathway analysis to test whether internal features play causal roles in reasoning outputs [Li et al., 2024]. We build on this intervention mindset but use a lightweight layer-ablation proxy on a small open model to keep the full protocol executable end to end.

Positioning. Our work sits between purely behavioral benchmarking and fine-grained mechanistic tracing. We pair real-model behavior (API evaluation) with internal interventions (open-model proxy) on aligned prompts and matched examples. The central outcome is a calibrated negative result: strong degradation and strong localization are both unsupported in this configuration.

3 Methodology

Hypothesis and design. We test whether commonsense failures primarily reflect breakdowns in goal-action coherence integration. The study has two tracks: (1) paired behavioral evaluation on an API model and (2) intervention-based mechanistic proxy analysis on an open model.

Dataset and sampling

We use the COMMONSENSEQA validation split from a local HuggingFace snapshot. The full split has 1,221 items; we deterministically shuffle with seed 42 and evaluate the first 120 items. Each item is formatted as a five-option multiple-choice prompt with labels A–E. Data checks verify complete fields, valid option sets, and valid gold labels.

Behavioral conditions and parsing

For each item, we query GPT-4.1 under two conditions at temperature 0.0:

Table 1: Behavioral accuracy on 120 COMMONSENSEQA validation items. Wilson 95% confidence intervals are reported. Best accuracy in **bold**.

Method	Accuracy	95% CI	N
DIRECT	**0.900**	[0.833, 0.942]	120
COT	0.892	[0.823, 0.936]	120

Table 2: Directional failure flips between conditions.

Flip type	Rate	Count
DIRECT correct → COT wrong	0.0250	3
COT correct → DIRECT wrong	0.0167	2

- DIRECT: short answer-only format (max tokens 16).
- COT: rationale-enabled format with explicit final tag `FINAL_ANSWER: <A-E>` (max tokens 200).

We parse final choices robustly and reject non A–E outputs (observed invalid rate: 0%).

Metrics and statistical tests

We report accuracy with Wilson 95% confidence intervals, directional flip rates (DIRECT →COT and COT →DIRECT), and McNemar exact test on discordant pairs. For coherence, we sample 80 items (160 condition-rows) and use GPT-4.1 as a judge to score whether the selected option is consistent with the likely question goal-action relation.

Mechanistic proxy intervention

We run a local LAYER-ABLATION PROXY in DISTILGPT2 (6 transformer layers). We construct matched success/failure examples from the DIRECT condition, compute base final-token logit margins,

$$\Delta_{\text{margin}} = \text{logit}(y_{\text{gold}}) - \max_{y' \neq y_{\text{gold}}} \text{logit}(y'), \tag{1}$$

and measure post-ablation change per layer as

$$\delta_\ell = \Delta_{\text{margin}}^{\text{ablated } \ell} - \Delta_{\text{margin}}^{\text{base}}. \tag{2}$$

We compare a prespecified target layer (layer 3) against control layers with paired t-test and Wilcoxon signed-rank test.

Implementation details

Key software versions are Python 3.12.8, openai 2.24.0, transformers 4.51.3, torch 2.7.1+cu126, scipy 1.17.1, and statsmodels 0.14.6. Hardware is 2× RTX 3090 (24GB). End-to-end runtime is approximately 8 minutes.

4 Results

Main behavioral comparison. Table 1 shows near-identical performance between DIRECT and COT conditions. DIRECT is slightly higher (90.0% vs. 89.2%), but confidence intervals overlap broadly.

McNemar exact test on discordant pairs gives $b = 3$, $c = 2$, $n_{\text{discordant}} = 5$, and $p = 1.0$, so we do not reject equal paired performance.

Coherence scoring. Coherence judgments are close to ceiling in most groups, which limits discriminative power for failure diagnosis. Figure 2 illustrates this saturation pattern.

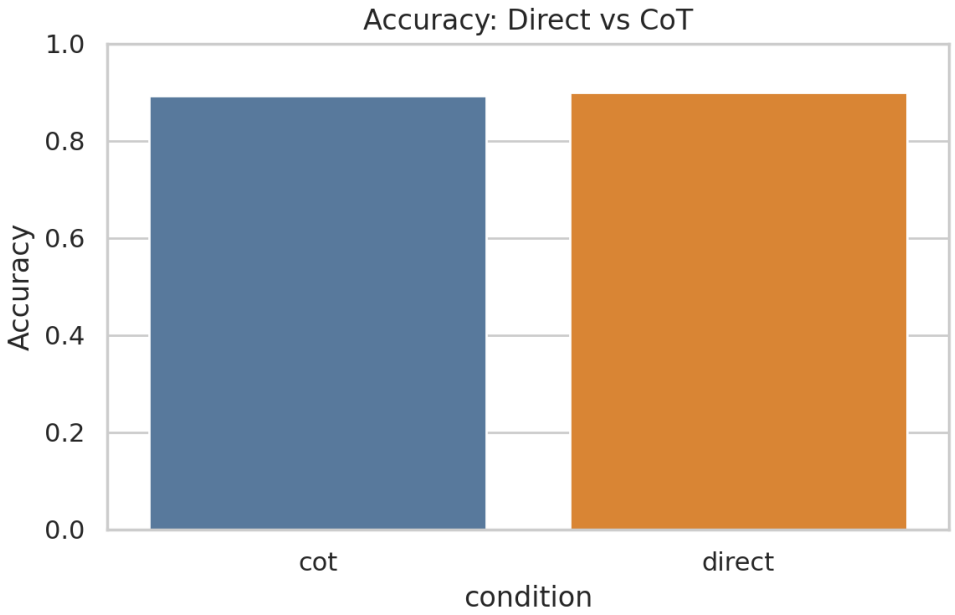

Figure 1: Behavioral accuracy comparison between DIRECT and COT prompting. Error bars visualize uncertainty intervals and show no practical gap.

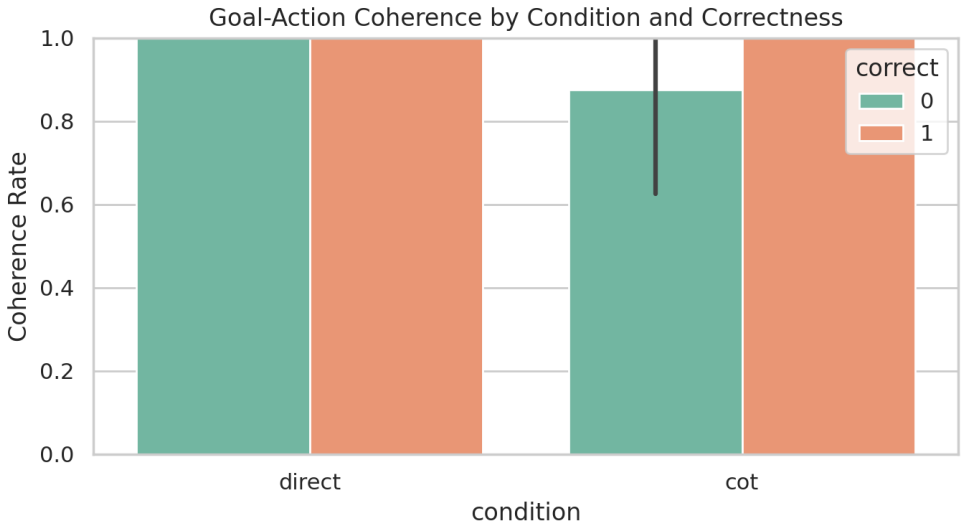

Figure 2: Goal-action coherence rates by condition and outcome subgroup. Rates cluster near 1.0, indicating a ceiling effect in this judge configuration.

Mechanistic proxy intervention results. Table 3 summarizes the layer-ablation proxy test. The target-layer effect is small and not significant by either parametric or nonparametric paired testing.

Ablation interpretation. The intervention track is a coarse ablation study at layer granularity. It does not isolate head-level or path-level mechanisms, and the null result should be interpreted as non-localization under this proxy rather than evidence that no mechanism exists.

Table 3: Mechanistic proxy test on matched success/failure examples in DISTILGPT2.

Statistic	Value
Matched examples	24
Layers scanned	6
Target layer	3
Paired t-test p	0.853
Wilcoxon p	0.546
Cohen's d (paired)	0.038

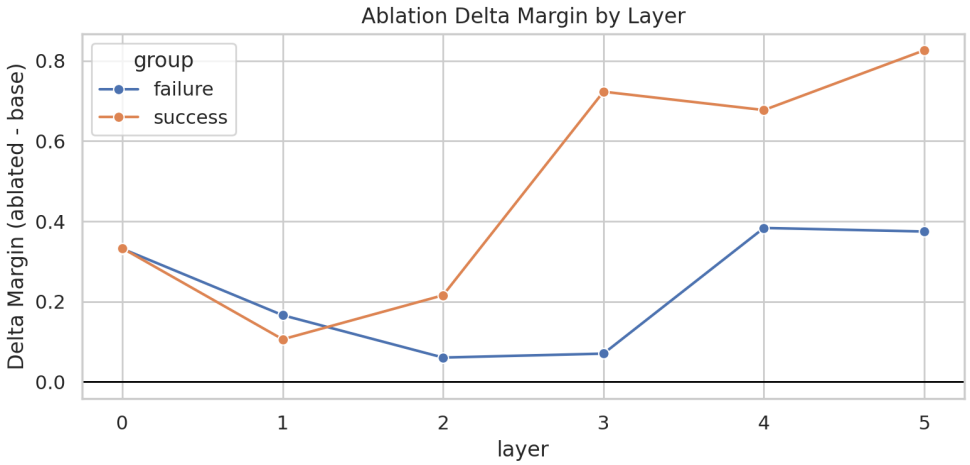

Figure 3: Layer-wise ablation deltas in coherent-margin proxy. Variability dominates and no single layer shows a robust directional effect in this small model scan.

5 Discussion

What the results support. The behavioral evidence is stable under paired comparison: CoT does not materially improve or degrade COMMONSENSEQA accuracy in this sampled setting. Flip rates are low in both directions, and the McNemar test indicates no significant difference. This weakens broad claims that CoT is generally toxic for commonsense QA.

What the results do not support. Our intervention proxy does not provide statistically significant localization of a coherence-related layer effect. Because the mechanistic track uses DISTILGPT2 instead of the API model, the result is best read as a construct-validity bound on this proxy setup, not a definitive mechanistic refutation.

Infrastructure lesson. An earlier run produced apparent extreme CoT degradation caused by a parsing bug, and the effect disappeared after parser correction. This is a direct reminder that evaluation pipelines can create false mechanism narratives if output extraction is not robust.

Limitations and threats to validity. The study uses one benchmark subset (120 COMMONSENSEQA items), one primary API model, one seed, and a small open-model proxy. Coherence judgments show a ceiling effect, limiting subgroup resolution. External validity to other datasets (e.g., Winogrande, ARC-Challenge) and other model families remains open. Internal validity is constrained by model mismatch between behavior and mechanism tracks.

Broader implications. Negative or null mechanistic findings are still informative when paired with rigorous behavioral controls. For reliability work, this suggests a staged standard: first establish robust paired behavior, then escalate to finer-grained interventions (head/path/feature) on models where internal access better matches the deployed system.

6 Conclusion

We presented a paired behavioral–mechanistic protocol to test whether commonsense failures in LLMs are dominated by goal-action coherence breakdowns. On 120 COMMONSENSEQA validation items, GPT-4.1 showed similar performance under DIRECT and COT prompting (90.0% vs. 89.2%), low bidirectional flip rates, and no significant paired difference. In a local DISTILGPT2 layer-ablation proxy, we found no significant target-layer causal effect.

The key takeaway is that strong claims about CoT-induced commonsense degradation are not supported in this sampled setting, and current small-model layer ablations provide weak localization evidence.

Future work should expand the paired protocol to additional benchmarks, replace ceiling-prone automatic coherence judges with validated annotation, and move from layer-level ablation to head-level activation patching or feature-level causal interventions in larger open models.

References

Vivien Cabannes, Aryaman Arora, Aadhitya Radhakrishnan, Neel Nanda, Anirudh Goyal, and Surya Chaudhuri. Iteration head: A mechanistic study of chain-of-thought. In *Advances in Neural Information Processing Systems*, 2024.

Yifan Hou, Jiacheng Zhang, Zhong Rao, and William Yang Wang. Towards a mechanistic interpretation of multi-step reasoning capabilities of language models. In *Proceedings of EMNLP*, 2023.

Jiachun Li et al. Focus on your question! interpreting and mitigating toxic cot problems in commonsense reasoning. *arXiv preprint arXiv:2402.18300*, 2024.